Worksheet - 03

Course: Aptitude **Topic**: Averages

1. The average of 10 numbers is 35. If one number is excluded, the average becomes 33. What is the excluded number?
   A) 45 B) 50 **C) 53** D) 65
   Sol:
   Sum of 10 numbers = 10 × 35 = 350
   Sum of remaining 9 numbers = 9 × 33 = 297
   Excluded number = 350 − 297 = 53

2. The average age of 30 students is 14 years. When a teacher joins, the average increases by 0.5 years. What is the age of the teacher?
   A) 29 **B) 29.5** C) 31.5 D) 32
   Sol:
   Original total age = 30 × 14 = 420
   New average = 14.5
   New total = 31 × 14.5 = 449.5
   Teacher's age = 449.5 − 420 = 29.5

3. The average weight of 20 boys is 60 kg and that of 10 girls is 50 kg. What is the average weight of the whole group?
   A) 55 kg **B) 56.7 kg** C) 57 kg D) 58 kg
   Sol:
   Total weight = (20 × 60) + (10 × 50) = 1700
   Total people = 30
   Average = 1700 ÷ 30 = 56.7 Kg

4. The average of 8 numbers is 25. If each number is increased by 4, the new average is
   A) 27 B) 28 **C) 29** D) 30
   Sol:
   Increase in average = 4
   New average = 25 + 4 = 29

5. The average marks of a student in 5 subjects is 68. If he scores 82 in the 6th subject, what is the new average?
   A) 70.33 B) 71 C) 7222 D) 73
   Sol:
   Total marks = 5 × 68 = 340
   New total = 340 + 82 = 422
   New average = 422 ÷ 6 = 70.33

6. The average of first 20 natural numbers is
   A) 9.5 B) 10 **C) 10.5** D) 11
   Sol:
   Average of first n natural numbers = (n + 1)/2
   = (20 + 1)/2 = **10.5**

7. The average age of 6 persons is 24 years. When a person aged 36 years joins, the new average becomes
   A) 25.71 B) 26 C) 27 D) 28

Sol:
Total age = 6 × 24 = 144
New total = 144 + 36 = 180
New average = 180 ÷ 7 =25.71

8. The average of 12 numbers is 40. If two numbers 30 and 50 are replaced by 35 and 45, the new average is
A) 39 **B) 40** C) 41 D) 42
Sol:
Old sum = 12 × 40 = 480
Removed sum = 30 + 50 = 80
Added sum = 35 + 45 = 80
No change in sum → Average = 40

9. The average of 15 observations is 22. If each observation is multiplied by 2, the new average becomes
A) 22 B) 33 **C) 44** D) 66
Sol:
New average = 22 × 2 = 44

10. The average of 9 numbers is 28. If one number is deleted, the average becomes 26. The deleted number is
A) 42 **B) 44** C) 46 D) 48
Sol:
Original sum = 9 × 28 = 252
New sum = 8 × 26 = 208
Deleted number = 252 − 208 = 44

11. The average monthly income of 10 persons is ₹20,000. If one person earns ₹35,000, the average income of the remaining persons is
A) ₹18,000 **B) ₹18,500** C) ₹19,000 D) ₹19,500
Sol:
Total income = 10 × 20000 = 200000
Remaining income = 200000 − 35000 = 165000
Average = 165000 ÷ 9 = ₹18,500

12. The average of 5 consecutive even numbers is 26. What is the largest number?
A) 30 **B) 32** C) 34 D) 36
Sol:
Middle number = 26
Numbers: 22, 24, 26, 28, 30
Largest = 30

13. A pupil's mark was wrongly entered as 83 instead of 63. Due to that, the average mark of the class increased by half a mark. The number of pupils in the class is:
A) 20 **B) 40** C) 30 D) 15
Sol:
The mark was wrongly entered as 83 instead of 63. So the excess mark is 20. It is said that this 20 marks has resulted in increase of ½ marks for each student. Therefore, 2 students together make 1 mark.
Let x be the number of students.
½*x = 20
x = 40

14. The average age of 15 students of a class is 15 years. Out of these, the average age of 5 students is 14 years and that of the other 9 students is 16 years, The age of the 15th student is**:**
A)12 **B) 11** C) 3 D) 15
Sol:
Age of the 15th student = [15 * 15 - (14 * 5 + 16 * 9)] = (225-214) = 11 years.

15. The average salary of 15 employees is ₹25,000. If the manager earning ₹55,000 is excluded, the new average salary is
A) ₹22,000 **B) ₹23,000** C) ₹24,000 D) ₹25,000
Sol:
Total salary = 15 × 25000 = 375000
Remaining salary = 375000 − 55000 = 320000
Employees = 14
Average = 320000 ÷ 14 ≈ ₹23,000

16. The average of 6 numbers is 18. If the average of 2 of them is 25, the average of the remaining numbers is
A) 14.5 **B) 15** C) 16.5 D) 17
Sol:
Total sum = 6 × 18 = 108
Sum of 2 numbers = 2 × 25 = 50
Remaining sum = 58
Average = 58 ÷ 4 = 14.5

17. The average speed of a car for a journey of 300 km is 60 km/h. The time taken is
A) 4 hours **B) 5 hours** C) 6 hours D) 7 hours
Sol:
Time = Distance ÷ Speed = 300 ÷ 60 = 5 hours

18. The average of 11 numbers is 50. If the average of the first 6 numbers is 45, the average of the remaining numbers is
A) 54 **B) 56** C) 57 D) 58
Sol:
Total sum = 11 × 50 = 550
First 6 sum = 6 × 45 = 270
Remaining sum = 280
Average = 280 ÷ 5 = 56

19. The average age of a family of 6 members is 28 years. If the youngest member is 4 years old, the average age of the remaining members is
A) 30 **B) 32** C) 34 D) 36
Sol:
Total age = 6 × 28 = 168
Remaining total = 168 − 4 = 164
Average = 164 ÷ 5 = 32

20. The average of 40 numbers is 25. If each number is decreased by 3, the new average is
A) 20 B) 21 **C) 22** D) 23
Sol:
New average = 25 − 3 = 22

21. The average marks of 40 students is 65. If the average of 25 students is 70, the average of the remaining students is **(CAPGEMINI)**
A) 55.66 **B) 56.66** C) 57 D) 58
Sol:

Total marks = 40 × 65 = 2600
First group = 25 × 70 = 1750
Remaining = 850
Average = 850 ÷ 15 =56.66

22. The average of 10 numbers is 50. If two numbers are removed, the average becomes 48. The average of the removed numbers is **(VIRTUSA)**
A) 56 **B) 58** C) 60 D) 62
Sol:
Original sum = 500
Remaining sum = 8 × 48 = 384
Removed sum = 116
Average = 116 ÷ 2 = 58

23. The average age of 12 persons is 30 years. If 4 persons leave whose average age is 24, the new average age is **(CTS)**
A) 32 **B) 33** C) 34 D) 35
Sol:
Total age = 12 × 30 = 360
Leaving = 4 × 24 = 96
Remaining = 264
Average = 264 ÷ 8 = 33

24. The average weight of a class of 25 students is 48kg. If the teacher's weight is included, the average increases by 2kg. Find the weight of the teacher? (**TCS)**
A)120 Kgs **B)100 Kgs** C)140 Kgs D)150 Kgs
Sol:
The average weight of a class of 25 students is 48 kg
=>25*48 = 1200
When teacher's weight is included average increases by 2, So
=>26*50 = 1300
=>1300-1200 = 100
The weight of the teacher is 100 kg.
Alternate method 25 students weight increases by 2 kg, so 2*25 = 50 and the weight of the teacher is 50 => 50+50 = 100 kg

25. The average of 60 numbers is 45. If two numbers, namely 59 and 60 are discarded, the average of remaining numbers is
A)44.5 B)45 C)40 D)48
Sol:
The average of 60 numbers is 45. Let us assume that each number will have 45 units.
Given, two numbers namely 59 and 60 are discarded.
We know that, the number of units with these discarded numbers would have also been 45.
So, the difference in units = 59-45 = 14 and 60-45 = 15 => 14+15 = 29
These 29 units have to be equally withdrawn from the remaining 58 numbers.
=>29/58 = 0.5
This 0.5 will be reduced from the initial average.
=>45-0.5 = 44.5

PRACTICE:

1. The average of 7 consecutive numbers is 35. What is the smallest number?
A) 29 B) 31 C) 32 D) 33

2. The average of 9 consecutive odd numbers is 41. What is the largest number?
A) 47 B) 49 C) 53 D) 55

3. The average expenditure of a man for 30 days is ₹120 per day. If he spends ₹150 per day for 20 days, his average expenditure for the remaining days is **(CTS)**
A) ₹80 B) ₹90 C) ₹100 D) ₹110

4. The average of 5 numbers is 18. If one number is replaced by 30, the new average becomes 20. The replaced number is **(INFOSYS)**
A) 18 B) 20 C) 22 D) 24

5. The average speed of a train for the first half of the journey is 60 km/h and for the second half is 40 km/h. The average speed of the entire journey is **(WIPRO)**
A) 48 km/h B) 50 km/h C) 52 km/h D) 54 km/h

6. The average age of 8 boys is 16 years. When a boy aged 23 leaves, the new average age is
A) 14.5 B) 15 C) 15.5 D) 16

7. The average of 4 numbers is 20. If three of them are 10, 15, and 25, the fourth number is
A) 30 B) 35 C) 40 D) 45

8. The average of 10 numbers is 35. If one number is added, the average becomes 36. The added number is
A) 45 B) 46 C) 47 D) 48

9. The average marks of a class of 50 students is 60. If 10 students with an average of 50 leave, the new average is
A) 61 B) 62.5 C) 63.5 D) 64

10. The average of 15 numbers is 22. If 5 numbers with an average of 30 are removed, the average of the remaining numbers is
A) 16 B) 17 C) 18 D) 19

11. The average income of A and B is ₹15,000. The income of A is ₹12,000. The income of B is
A) ₹16,000 B) ₹17,000 C) ₹18,000 D) ₹19,000

12. The average of 20 numbers is 18. If the average of the first 10 numbers is 15, the average of the remaining numbers is
A) 19 B) 20 C) 22 D) 21

13. The average of 11 numbers is 40. If one number is excluded, the average becomes 38. The excluded number is
A) 58 B) 64 C) 62 D) 60

14. The average of 6 numbers is 8. The average of another 4 numbers is 12. The average of all 10 numbers is **(CAPGEMINI)**
A) 9.6 B) 9.8 C) 10 D) 10.4

15. The average of 10 numbers is 50. If each number is multiplied by 2 and then reduced by 10, the new average is
A) 80 B) 85 C) 90 D) 95

16. The average of 5 numbers is 27. If four numbers are 25, 30, 35, and 20, the fifth number is
A) 20 B) 25 C) 30 D) 35

17. The average of 7 numbers is 31. If one number is excluded, the average becomes 28. The excluded number is
A) 43 B) 45 C) 47 D) 49

18. The average age of 5 children is 10 years. After 5 years, the new average age is
A) 12 B) 13 C) 14 D) 15

19. The average of 9 numbers is 45. What is the sum of the numbers?
A) 360 B) 380 C) 400 D) 405

20. The average of 12 numbers is 30. If the average of 8 numbers is 28, the average of the remaining numbers is **(TCS)**
A) 32 B) 33 C) 34 D) 35

21. The average of 4 numbers is 14. If the average of three numbers is 10, the fourth number is
A) 24 B) 26 C) 28 D) 30

22. The average salary of a group of workers is ₹18,000. If one worker earning ₹30,000 joins, the average becomes ₹19,000. The number of workers originally was
A) 8 B) 9 C) 10 D) 11

23. The average of 8 numbers is 25. If one number is replaced by 41, the average increases by 2. The original number was
A) 21 B) 23 C) 25 D) 27

24. The average of 50 numbers is 40. If 10 numbers with an average of 30 are removed, the average of the remaining numbers is **(CTS)**
A) 42.5 B) 43 C) 44.5 D) 45

25. The average marks of 60 students is 75. If 10 students with an average of 60 leave, the new average is
A) 77 B) 78 C) 79 D) 80

KEY:

1. B	11. B	21.B
2. C	12. D	22.D
3. B	13. D	23.C
4. B	14. A	24.A
5. B	15. C	25.B
6. C	16. B	
7. A	17. D	
8. B	18. D	
9. B	19. B	
10. C	20. C	